

Battery degradation

Model-based prediction

Final project for IronHack

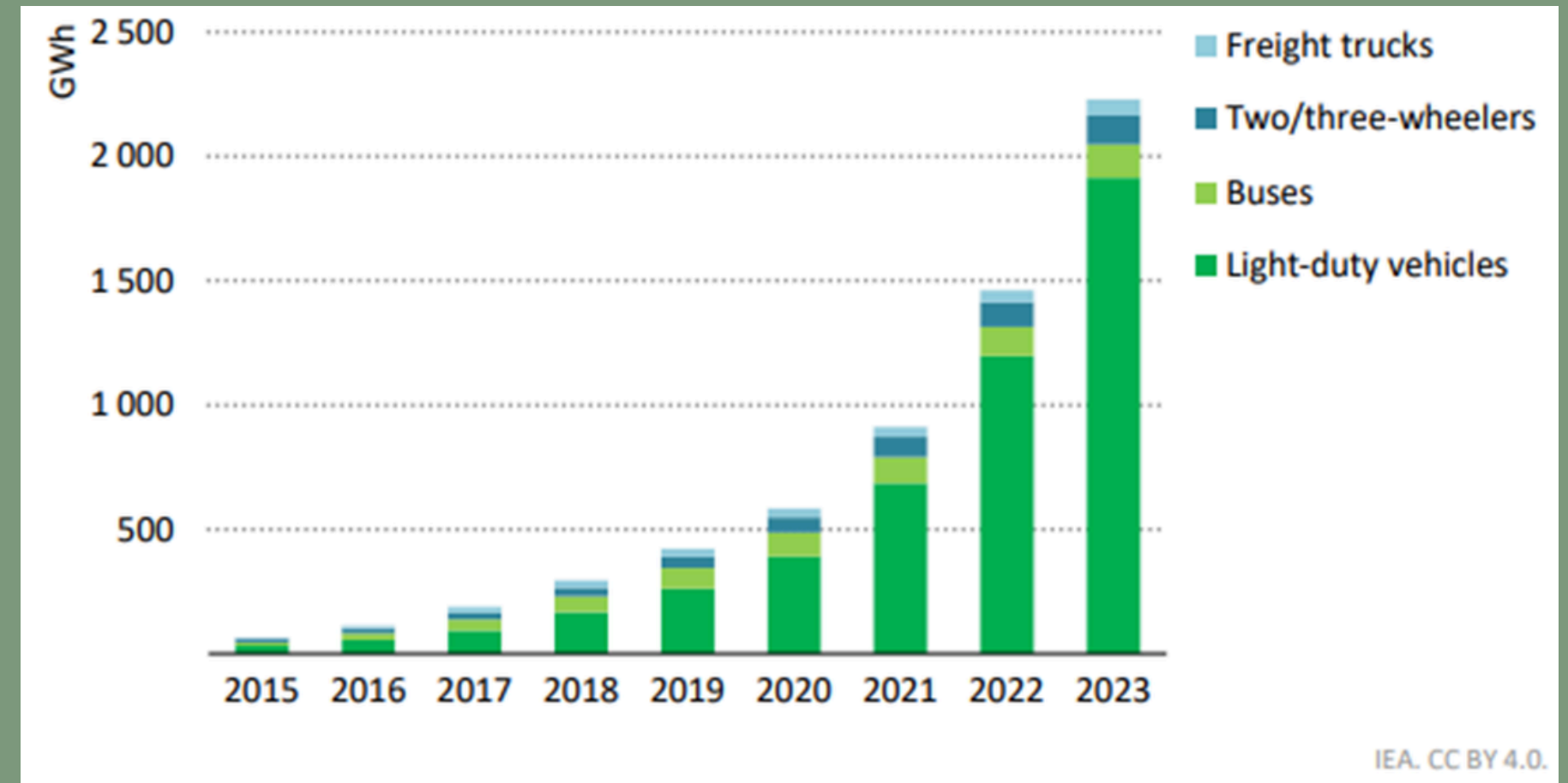
Nicole Segura Salas

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Rising demand of Li-ion batteries

Li-ion batteries are key for clean energy transition

- Increased government funding for EV research and infrastructure.
- It is essential to maximize and ensure safety operation

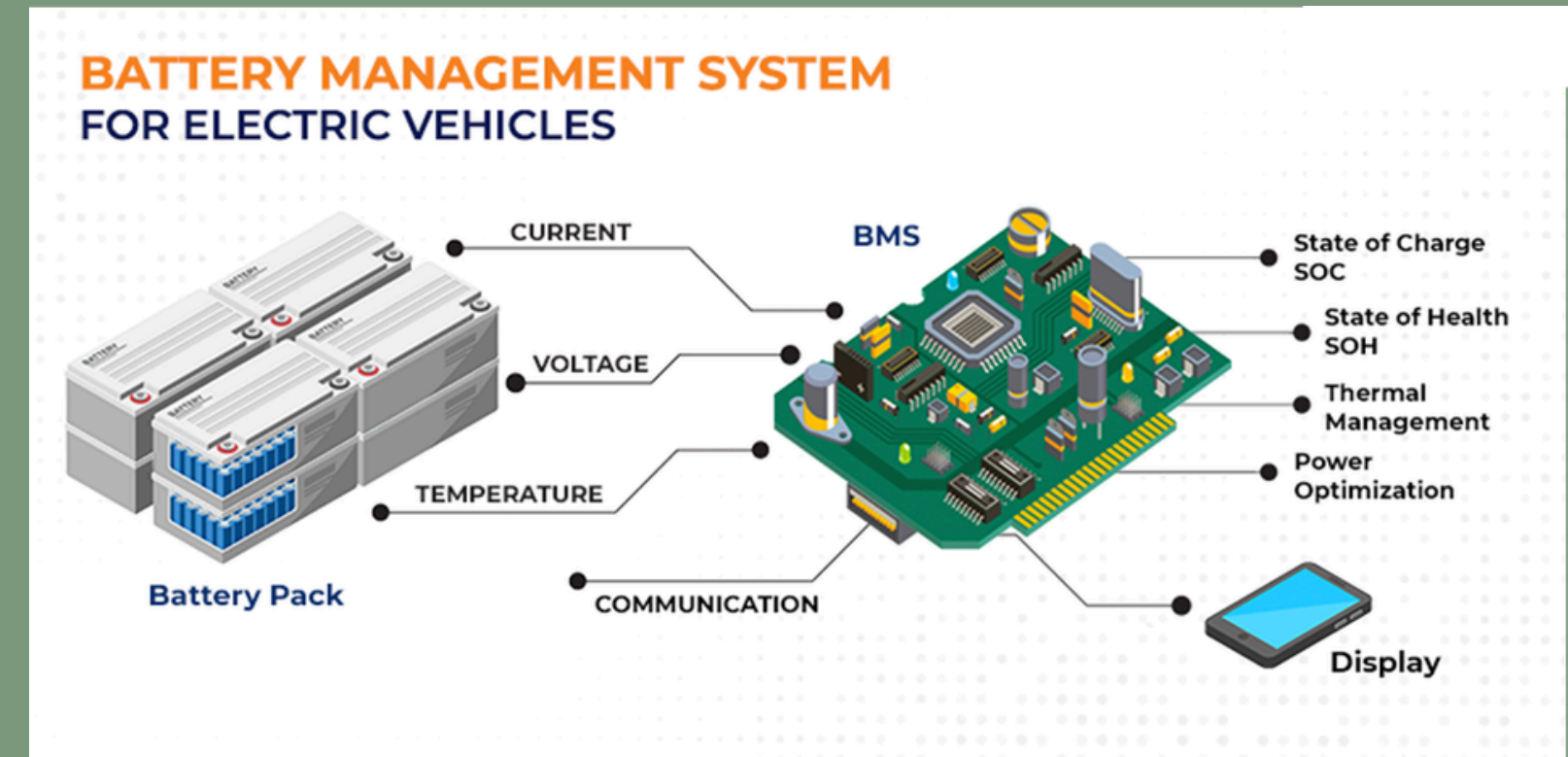


How to measure the battery health?

State of health (SoH):

Reflects the current state of the battery with respect to the manufacturer's values.

The nominal capacity is established by 2 Ah



$$SoH = \frac{\text{Current actual capacity (Ah)}}{\text{Nominal Capacity (Ah)}} \times 100$$

Dataset

Available in Kaggle

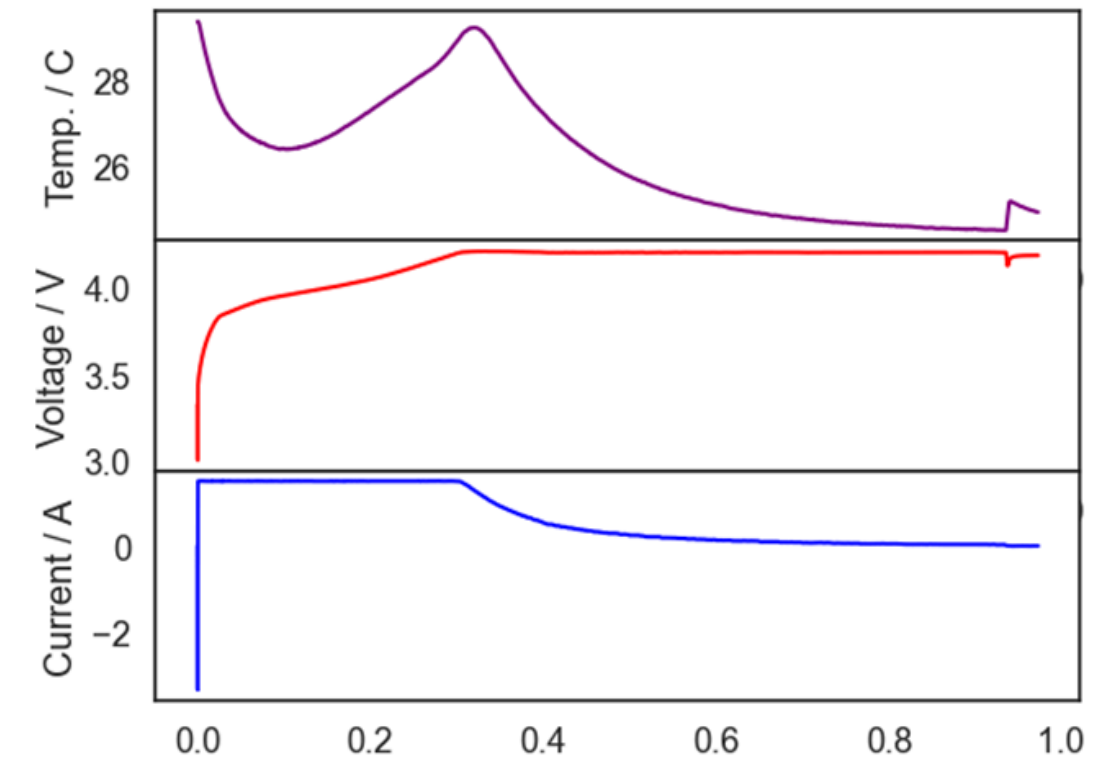
- NASA-battery dataset

Data structure

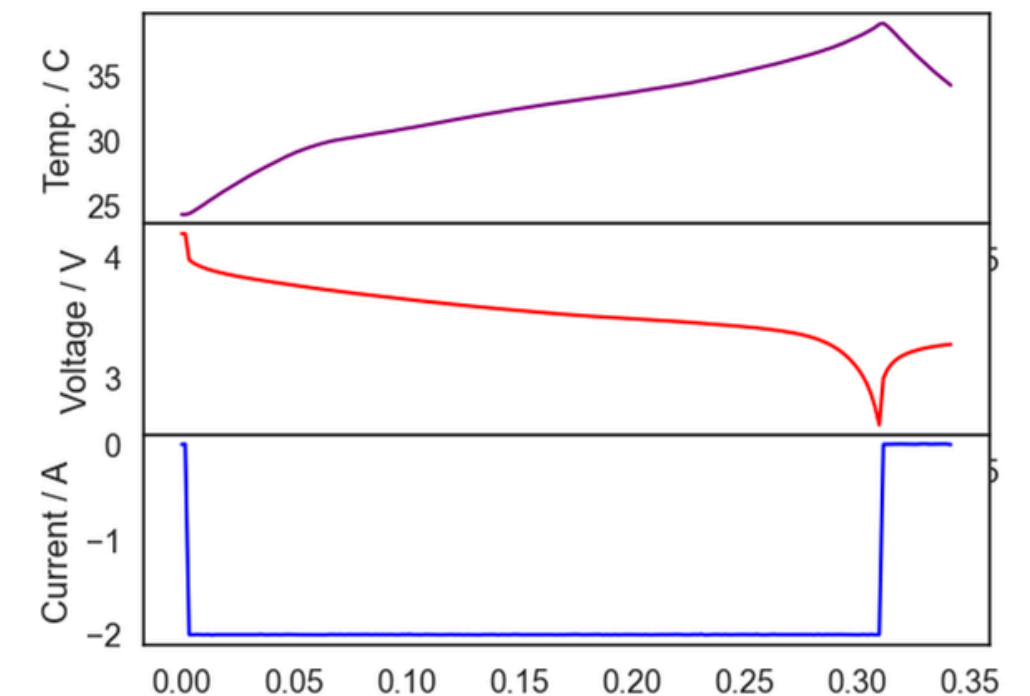
- 9 x 4 Li-ion batteries.
- Each was cycled through 3 different tests:
 - Charge
 - Discharge
 - Impedance
- At different temperatures
- Different current loads

Only 4 batteries were used during this project: #5-8 treated at 24 °C

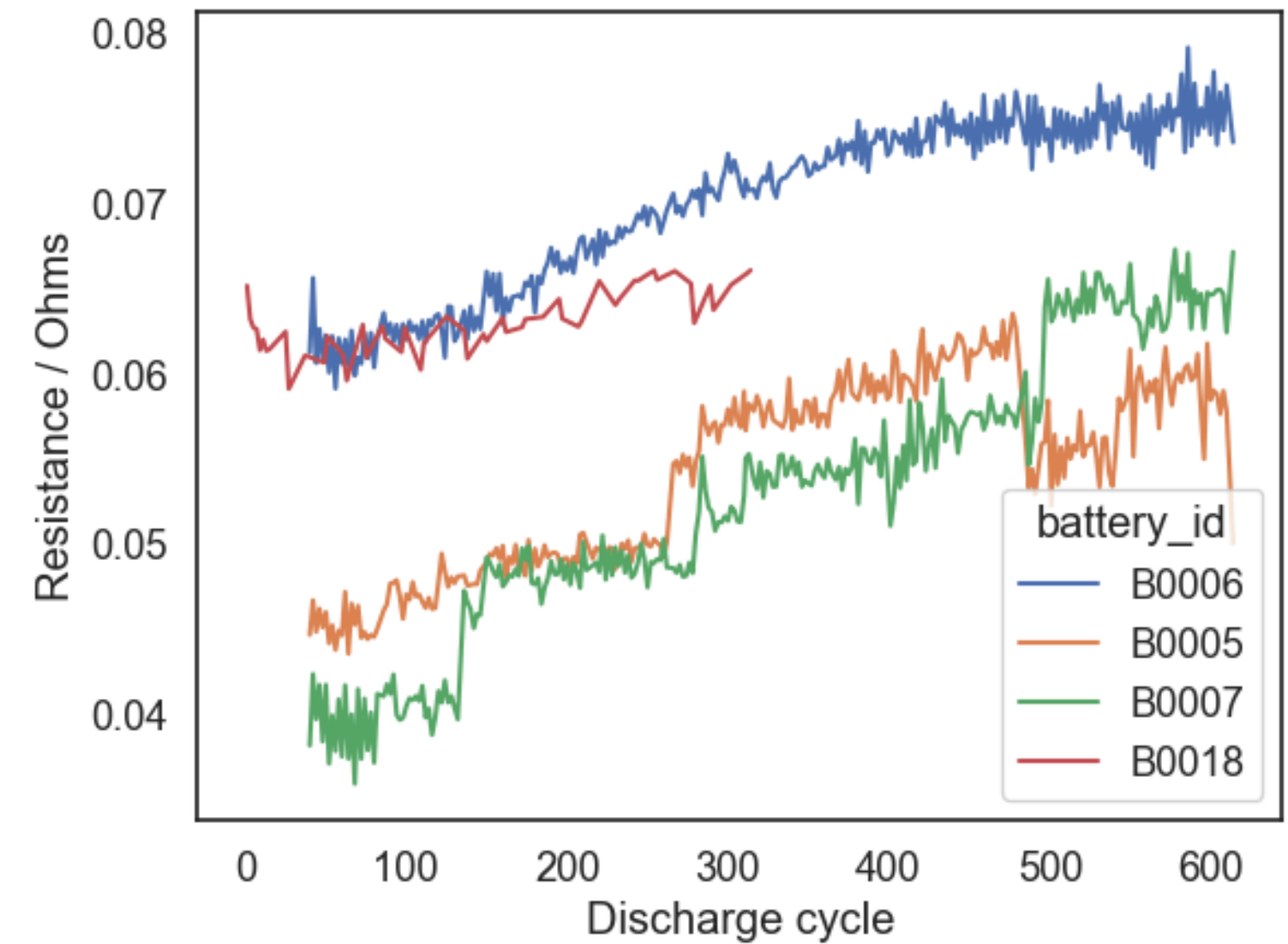
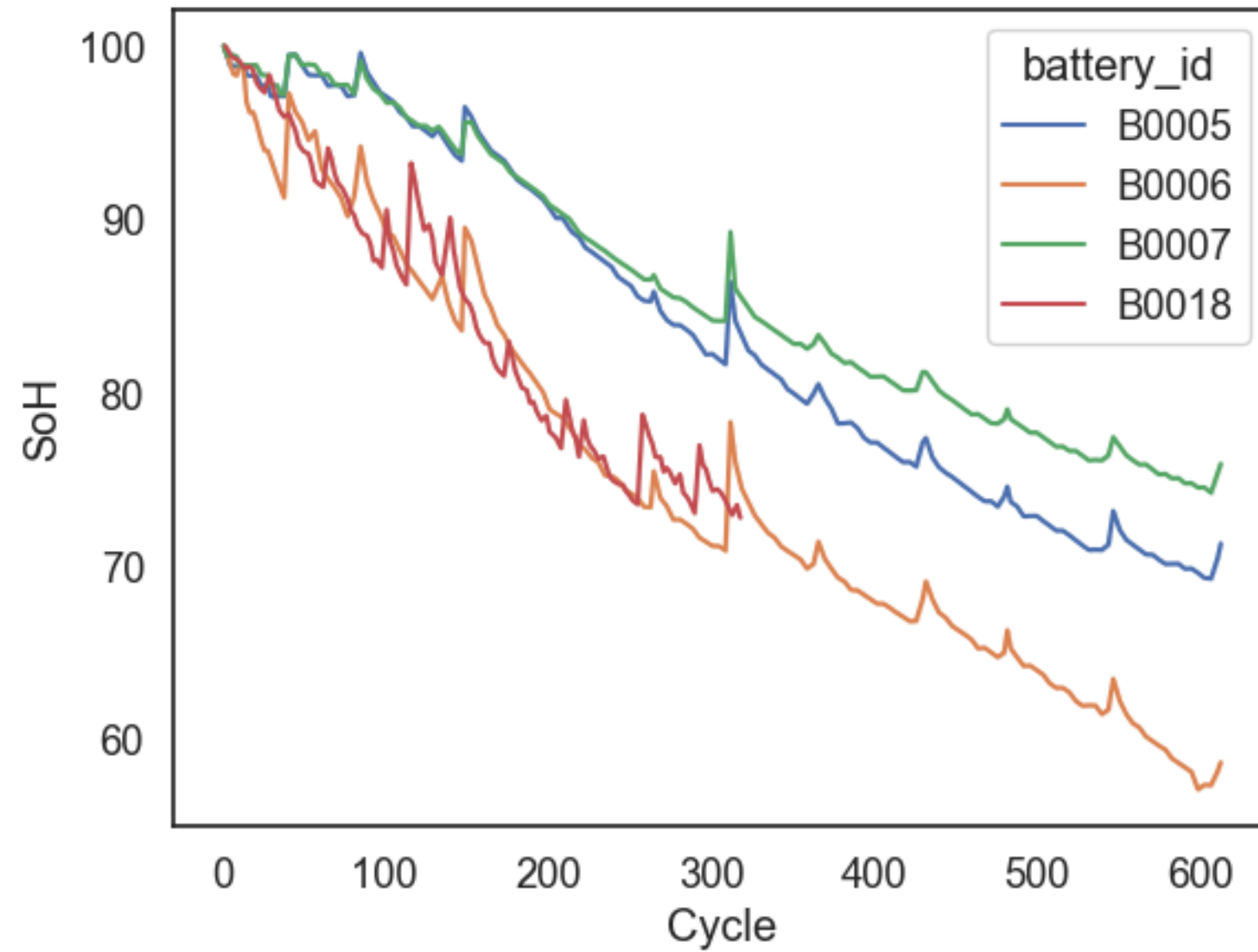
Charging cycle



Discharging cycle

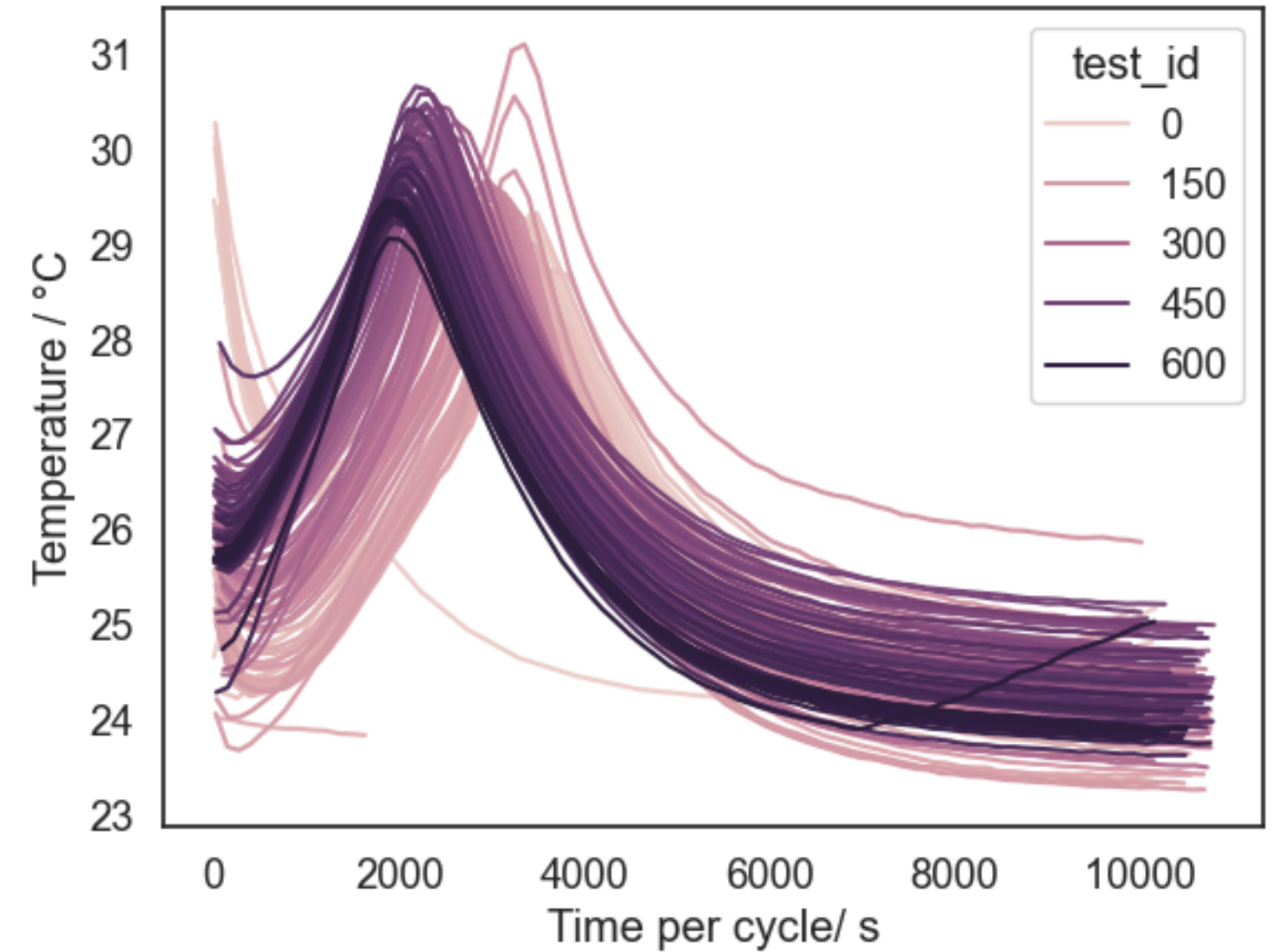
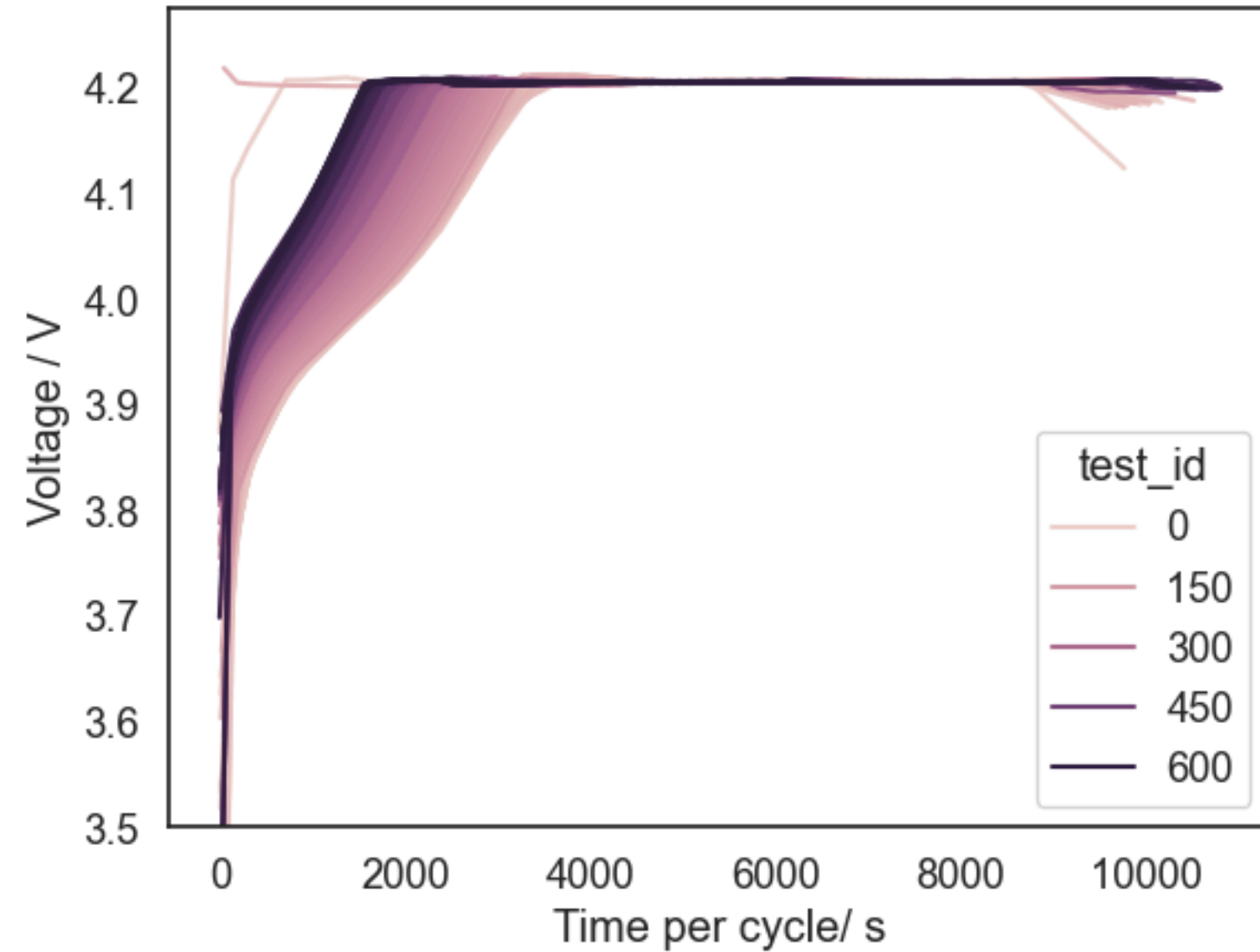


Degradation patterns

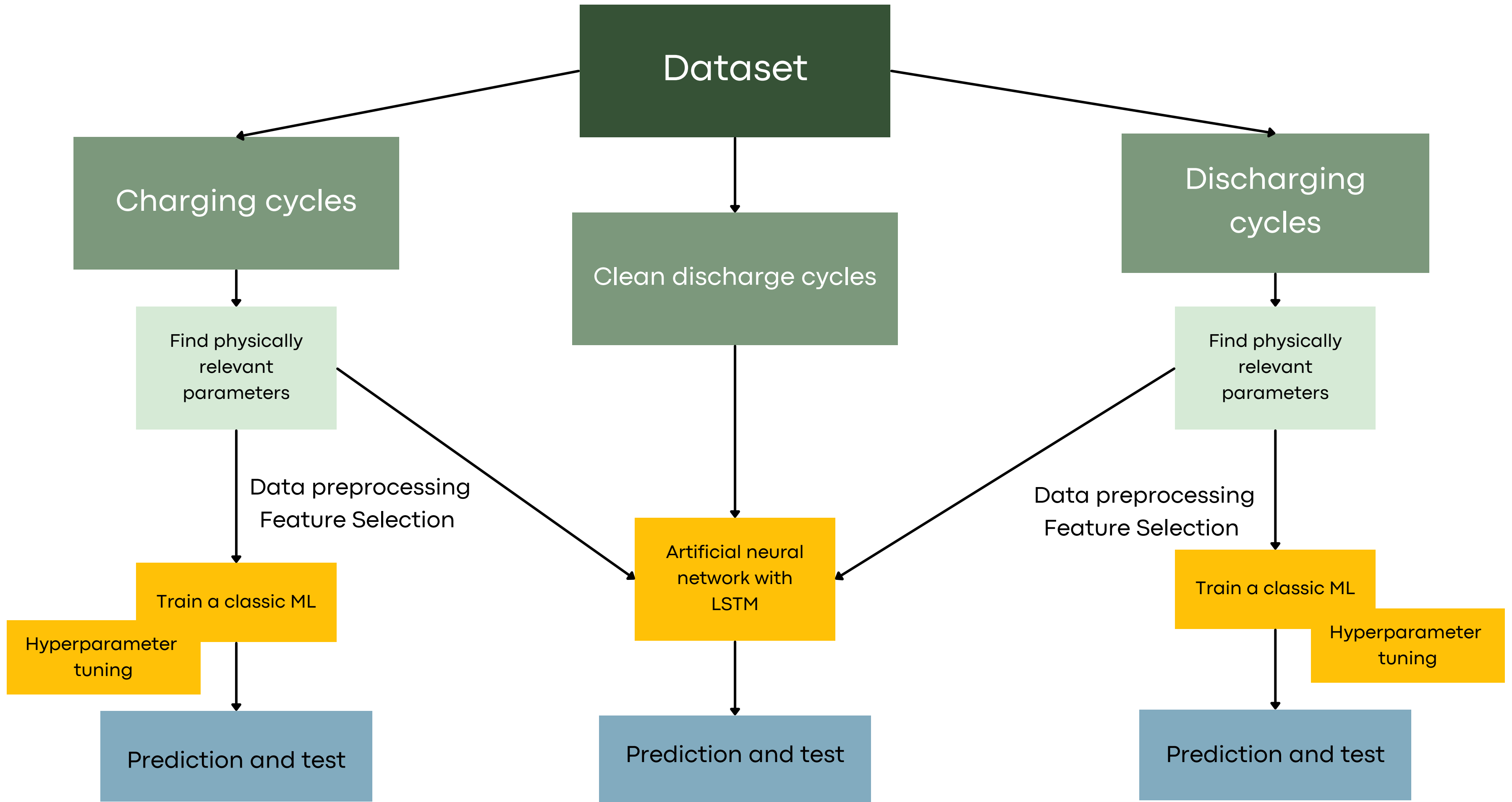


- General trend: Decrease in capacity output and increase in resistance
- Data is noisy and non-linear

Degradation patterns (Charging)



- Degradation is correlated to multiple parameters that relate to different battery behaviors



Models of choice

01

Xgboost

Decision tree-based.
Good at handling non-linear
relations
Easy to extract feature
importances

Requires feature engineering
Can't forecast
May overfit on small datasets

02

Supporting vector regressor

Handles well non-linear
relations

Requires feature engineering
Sensitive to scaling

03

LSTM

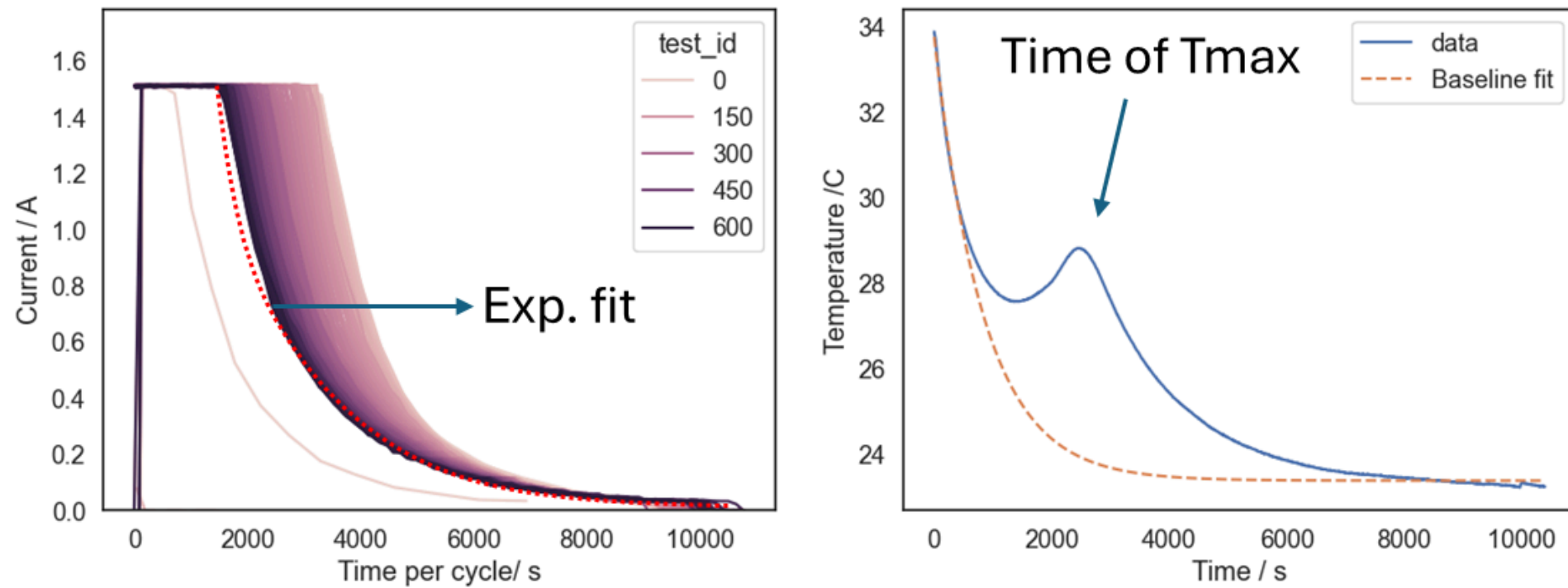
Is ideal for forecasting.
Retains history of previous
cycles

Requires large sets of data
Easily overfits
Higher computational resources
Difficult data preprocessing

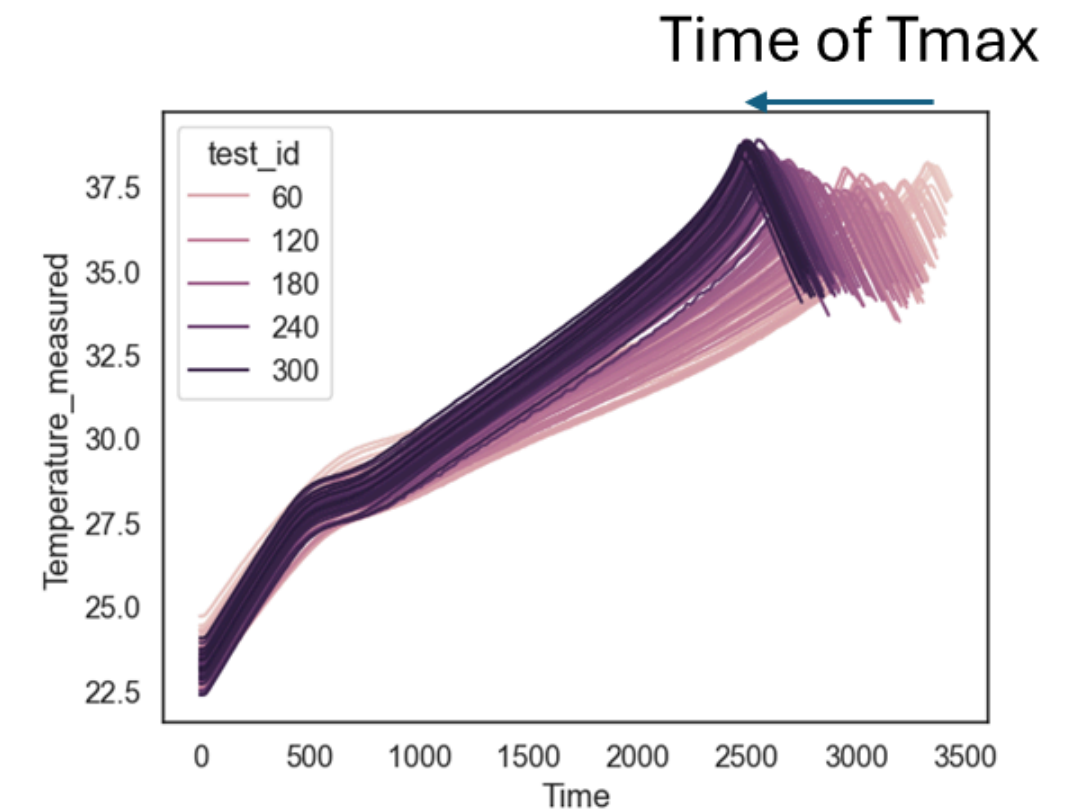
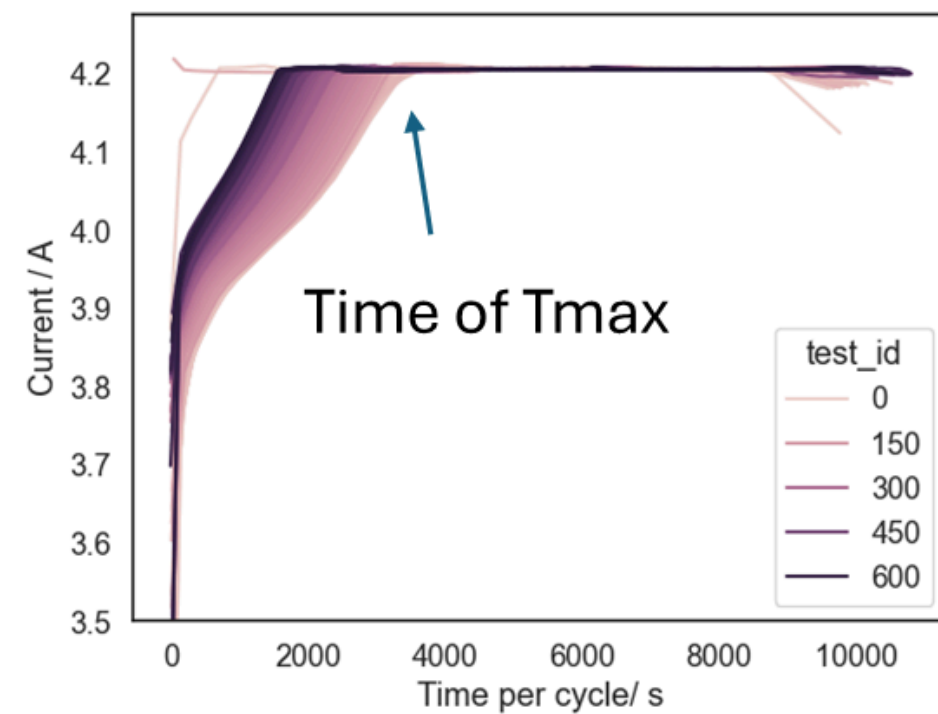
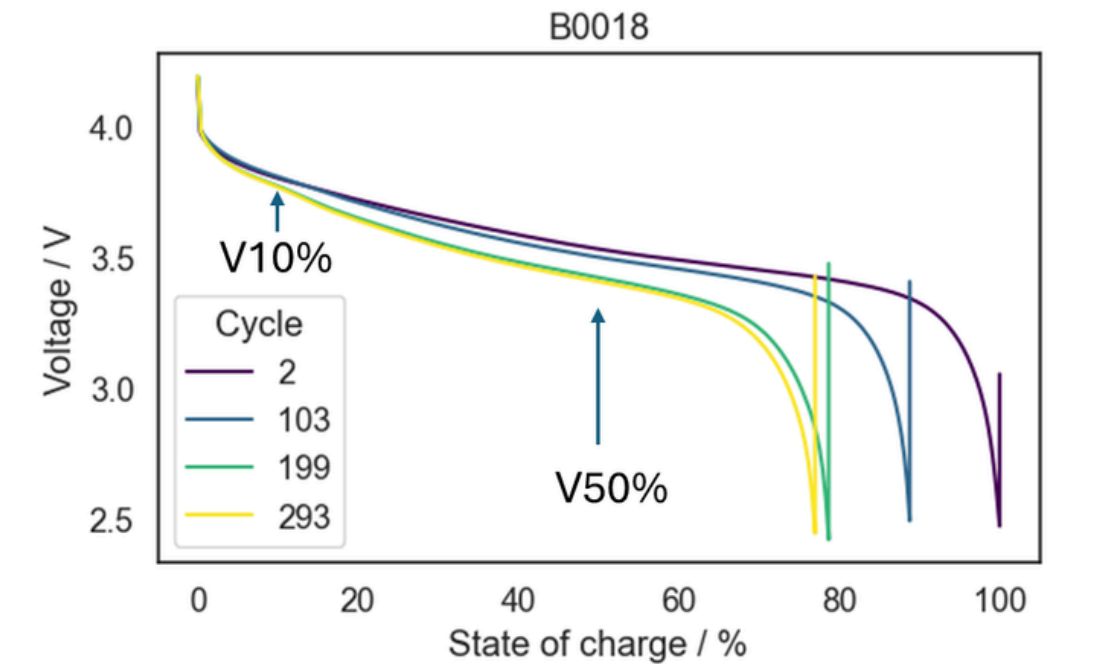
Feature engineering

After feature importance

Charging features



Discharging features

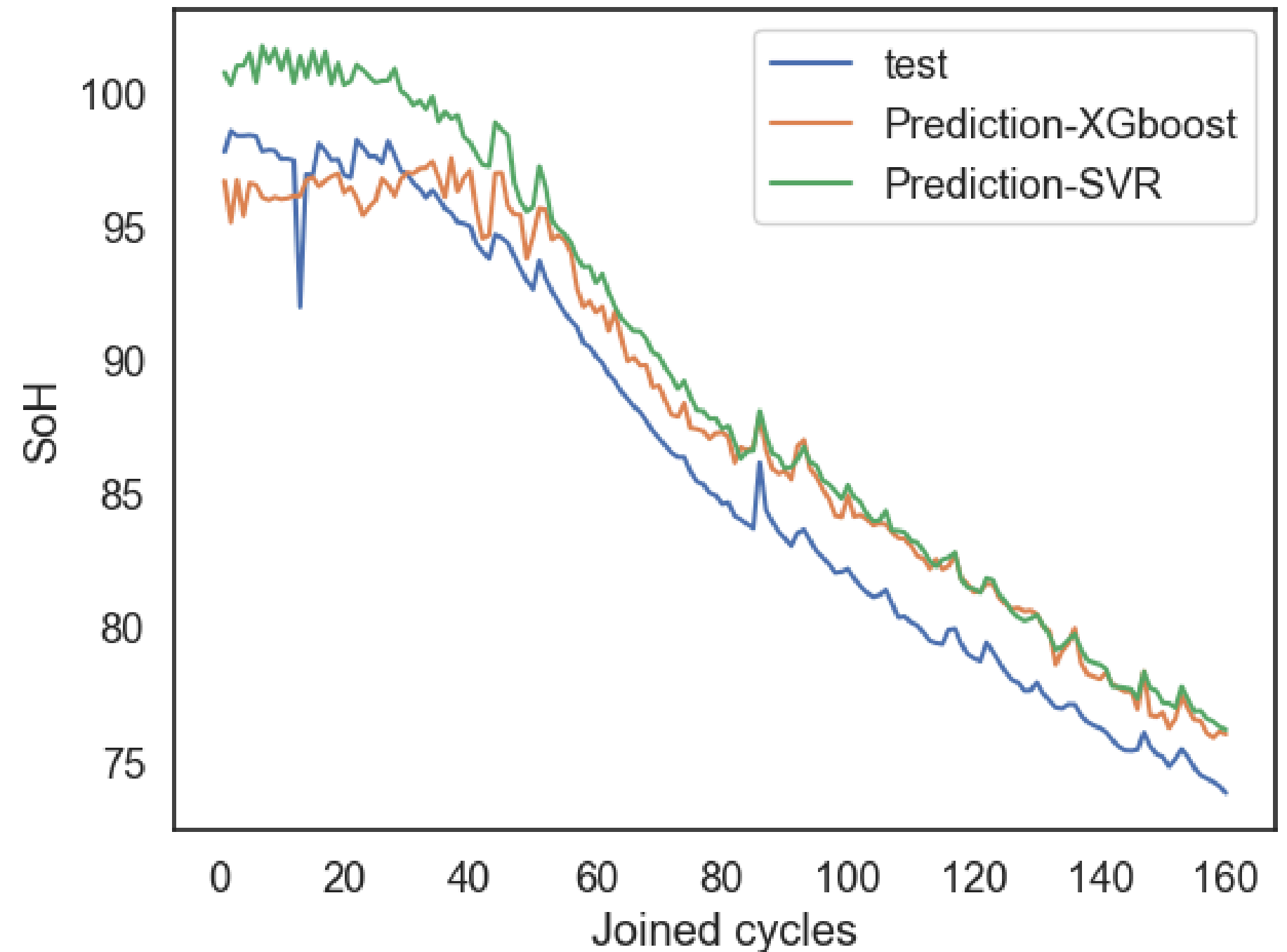


Results on static ML models

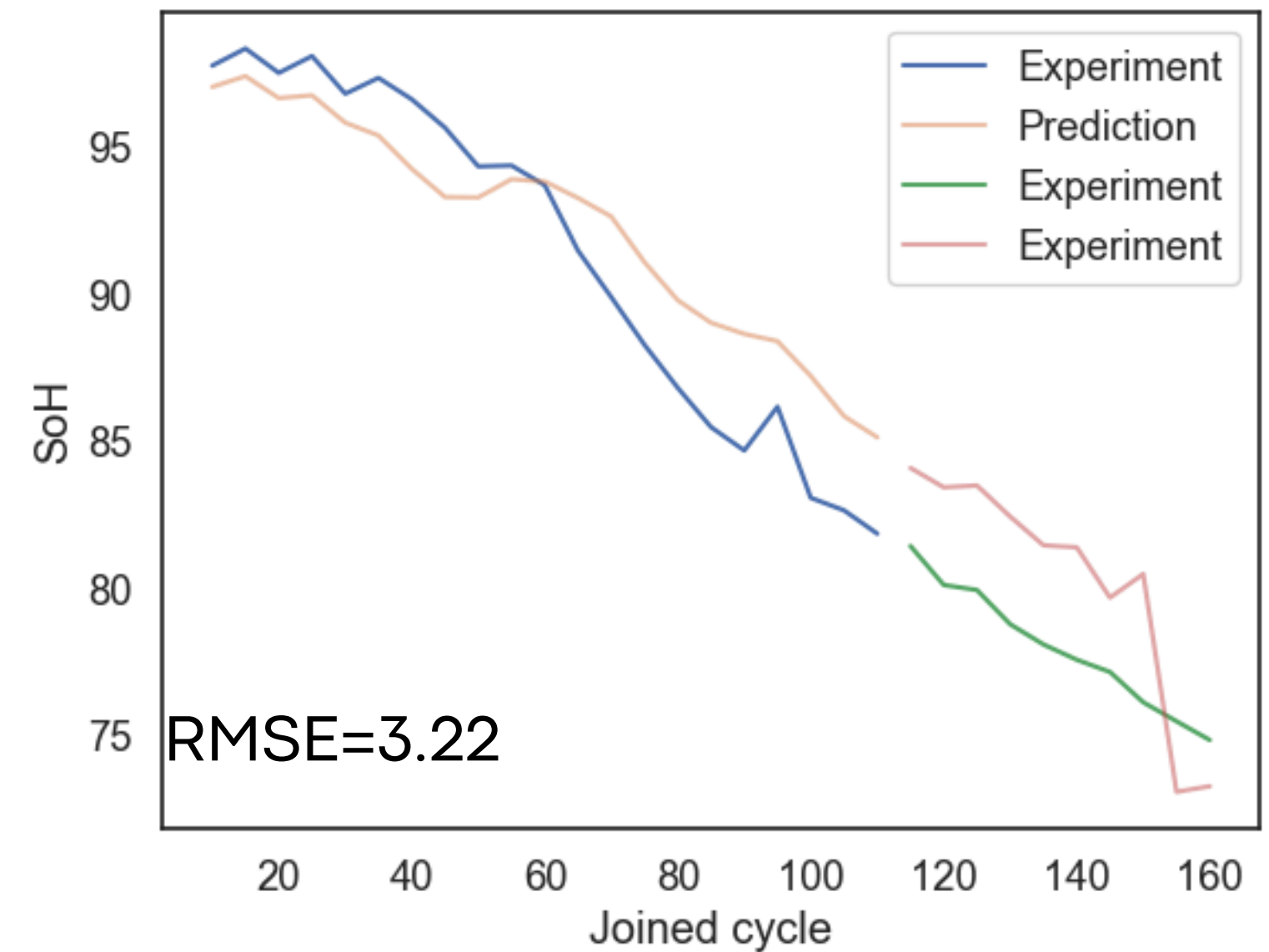
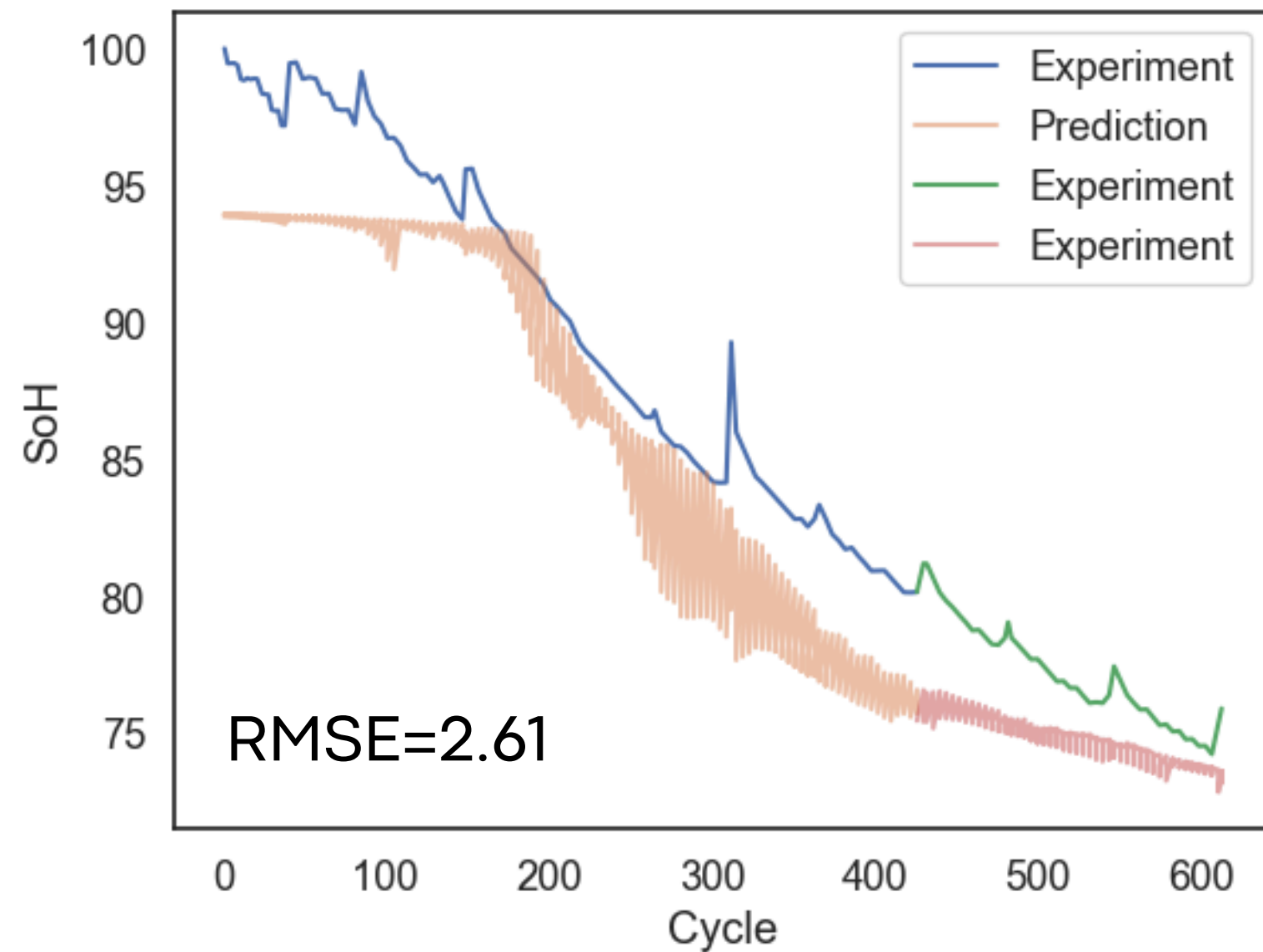
Main challenges:

- Generation of functions for feature extraction
- Dealing with outliers

	r2-score	RMSE
Static ML with XGBoost	0.923/0.88	2.34/2.88
Static SVR	0.948/0.905	1.92/2.61
Fused static ML XGboost	0.932	1.408
Fused static ML SVR	0.867	2.96



LSTM: Importance of feature engineering



LSTM requires the data to be in timesteps

Conclusions

Battery degradation is a time dependent process.

Feature engineering is essential to understand complex parameters

Data-driven models can accurately predict the SoH of a battery. Xgboost outperformed the use of SVR and LSTM

Future outlook: improve LSTM model architecture and include accelerating tests

**Thank you for your
attention!**